



United States ENVIRONMENTAL PROTECTION AGENCY

Washington, DC 20460

Laboratory DMR-QA Evaluation Study 31

Laboratory Performance Evaluation

Office of Enforcement and Compliance Assurance

(These data are collected under the authority of the Federal Water Pollution Control Act.)

Permittee name Cadillac WWTP State MI NPDES Permit No. 0020257 Permit Extension

Identification of all CHEM, MICRO and WET laboratories who did analyses for this permit

Name of Laboratory	Address of Laboratory	U.S. EPA Lab Code	Lab Analysis Check box(es) that apply			Lab Type*	State-certified Laboratory**
			Chem	Micro	WET		
Cadillac WWTP	200 N. Lake St Cadillac, MI 49601	MI00082	☒	☒	☐	G	☐
Kar Laboratories	4405 Manchester Rd Kalamazoo, MI 49001	MI00041	☒	☐	☐	C	☐
Paragon Laboratories	12649 Richfield Ct. Livonia, MI 48150	MI 01187	☐	☐	☒	C	☐
			☐	☐	☐		☐
			☐	☐	☐		☐
			☐	☐	☐		☐

* Lab Types: C = Commercial F = Federal G = Local Government I = Industrial O = Other S = State

** See Footnote 2 on DMR-QA Study 31 Fact Sheet

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If you need additional space, please make a copy of this page for additional laboratories.



United States ENVIRONMENTAL PROTECTION AGENCY

Washington, DC 20460

Laboratory DMR-QA Evaluation Study 31

Laboratory Performance Evaluation

Office of Enforcement and Compliance Assurance

(These data are collected under the authority of the Federal Water Pollution Control Act.)

NPDES Permittee Data Report Form**Due August
19, 2011****Attention:** Follow the instructions on the previous page to complete this form and submit data for evaluation.

State

NPDES Permit
Number

Permit Extension

MI

0020257

Permittee name

Cadillac WWTP

Current Permittee mailing address

200 N. Lake St

City

Cadillac

State

MI

Zip Code

49601

Phone Number

(231) 775-2841

FAX Number

(231) 775-2876

e-Mail

lab@cadillac-mi.net

For DMR-QA Study 31, conducted in 2011, the Permittee ensured that their laboratory(s) performing the required analyses:

Received PT Samples

Submitted Complete and Accurate Data
by July 1, 2011Received a Graded Report by
July 22, 2011Yes ☒ No ☐Yes ☒ No ☐Yes ☒ No ☐**Certification by Permit Holder or Authorized Representative**

(as per 40 C.F.R. Section 122.22 - see instructions.)

I certify under penalty of law that this document and all attachments were prepared under my direction or supervision in accordance with a system designed to assure that qualified personnel properly gather and evaluate the information submitted. Based on my inquiry of the person or persons who manage the system, or those persons directly responsible for gathering the information, the information submitted is, to the best of my knowledge and belief, true, accurate, and complete. Each reported value was produced from a single analytical run using the analytical system that routinely performs these analyses to produce compliance monitoring data required under our National Pollutant Discharge Elimination System (NPDES) permit. Neither I nor any of my subordinates compared our results with results from independent analyses conducted by us or any other laboratory before we reported our results to the U.S.EPA. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment for knowing violations.

Name of Certifying Official

Amy Vail

Title

Lab Manager

Signature

Amy Vail

Date

8/3/11

Address, phone number and e-mail of certifying official are required if different from above.

Address

200 N. Lake St

Phone No.

(231) 775-2368

City

Cadillac

State

MI

Zip Code

49601

E-mail

lab@cadillac-mi.net

Permittee name

Cadillac

State

MI

NPDES Permit No.

0000257

EPA Lab Code

MI00082

Chemistry/Microbiology Analyte Checklist

DMR-QA Study 31

Analyte Test	Test Required	Laboratory's Graded Result		Analyte determined by state-certified laboratory*
		Acceptable	Not Acceptable (Corrective Action Required)	
Microbiology				
E. coli	☐	☐	☐	☐
Fecal Coliform, MF or MPN	☒	☒	☐	☐
Total Coliform, MF or MPN	☐	☐	☐	☐
Trace Metals				
Aluminum	☐	☐	☐	☐
Antimony	☐	☐	☐	☐
Arsenic	☐	☐	☐	☐
Barium	☐	☐	☐	☐
Beryllium	☐	☐	☐	☐
Cadmium	☐	☐	☐	☐
Chromium, total	☐	☐	☐	☐
Chromium, hexavalent	☐	☐	☐	☐
Cobalt	☐	☐	☐	☐
Copper	☐	☐	☐	☐
Iron	☐	☐	☐	☐
Lead	☐	☐	☐	☐
Manganese	☐	☐	☐	☐
Mercury	☐	☐	☐	☐
Mercury (Low Level)	☐	☐	☐	☐
Molybdenum	☐	☐	☐	☐
Nickel	☐	☐	☐	☐
Selenium	☒	☒	☐	☐
Silver	☐	☐	☐	☐
Thallium	☐	☐	☐	☐
Vanadium	☐	☐	☐	☐
Zinc	☒	☒	☐	☐
Demands				
5-day BOD	☐	☐	☐	☐
5-day Carbonaceous BOD	☒	☒	☐	☐
COD	☐	☐	☐	☐
TOC	☐	☐	☐	☐
Minerals				
Alkalinity, total (CaCO ₃)	☐	☐	☐	☐
Chloride	☐	☐	☐	☐
Fluoride	☐	☐	☐	☐
Hardness, total (CaCO ₃)	☐	☐	☐	☐
Specific conductance (25°C)	☐	☐	☐	☐
Sulfate	☐	☐	☐	☐
Total Dissolved Solids (180°C)	☒	☒	☐	☐
Nutrients				
Ammonia as N	☒	☒	☐	☐
Nitrate as N	☐	☐	☐	☐
Nitrite as N	☐	☐	☐	☐
Orthophosphate as P	☐	☐	☐	☐
Total Kjeldahl-Nitrogen as N	☐	☐	☐	☐
Total Phosphorus as P	☒	☐	☒	☐
Misc. Analytes				
Non-Filterable Residue (TSS)	☒	☒	☐	☐
Oil and Grease	☐	☐	☐	☐
pH	☒	☒	☐	☐
Total Cyanide	☐	☐	☐	☐
Total Phenolics (4-AAP)	☐	☐	☐	☐
Total Residual Chlorine	☐	☐	☐	☐
Total Residual Chlorine (Low Level)	☐	☐	☐	☐
Settleable Solids	☐	☐	☐	☐
Turbidity	☐	☐	☐	☐

Signature

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Date

8/3/11

*See Footnote 2 on DMR-QA Study 31 Fact Sheet

WET Organisms/Test Conditions/End Points Checklist

DMR-QA Study 31

Analyte Number	Organisms / Conditions	End Points	Test Required	Laboratory's Graded Result		Analyte determined by state-certified laboratory*
				Acceptable	Not Acceptable (Corrective Action Required)	
Test Code 13/EPA Method 2000						
4	Fathead minnow (<i>Pimephales promelas</i>) - MHSF 25°C	LC50	☐	☐	☐	☐
Test Code 14/EPA Method 2000						
5	Fathead minnow (<i>Pimephales promelas</i>) - 20% DMW	LC50	☐	☐	☐	☐
Test Code 15/EPA Method 1000						
6	Fathead minnow (<i>Pimephales promelas</i>) - MHSF	NOEC SURVIVAL	☐	☐	☐	☐
8	Fathead minnow (<i>Pimephales promelas</i>) - MHSF	IC25 (ON) GROWTH	☐	☐	☐	☐
0	Fathead minnow (<i>Pimephales promelas</i>) - MHSF	NOEC (ON) GROWTH	☐	☐	☐	☐
Test Code 16/EPA Method 1000						
9	Fathead minnow (<i>Pimephales promelas</i>) - 20% DMW	NOEC SURVIVAL	☐	☐	☐	☐
2	Fathead minnow (<i>Pimephales promelas</i>) - 20% DMW	IC25 (ON) GROWTH	☐	☐	☐	☐
4	Fathead minnow (<i>Pimephales promelas</i>) - 20% DMW	NOEC (ON) GROWTH	☐	☐	☐	☐
Test Code 19/EPA Method 2002						
4	<i>Ceriodaphnia dubia</i> - MHSF 25°C	LC50	☒	☒	☐	☐
Test Code 20/EPA Method 2002						
5	<i>Ceriodaphnia dubia</i> - 20% DMW 25°C	LC50	☐	☐	☐	☐
Test Code 21/EPA Method 1002						
5	<i>Ceriodaphnia dubia</i> - MHSF	NOEC SURVIVAL	☒	☒	☐	☐
7	<i>Ceriodaphnia dubia</i> - MHSF	IC25 REPRODUCTION	☒	☒	☐	☐
8	<i>Ceriodaphnia dubia</i> - MHSF	NOEC REPRODUCTION	☒	☒	☐	☐
Test Code 22/EPA Method 1002						
1	<i>Ceriodaphnia dubia</i> - 20% DMW	NOEC SURVIVAL	☐	☐	☐	☐
1	<i>Ceriodaphnia dubia</i> - 20% DMW	IC25 REPRODUCTION	☐	☐	☐	☐
	<i>Ceriodaphnia dubia</i> - 20% DMW	NOEC REPRODUCTION	☐	☐	☐	☐
Test Code 32/EPA Method 2021						
	<i>Daphnia magna</i> - MHSF 25°C	LC50	☐	☐	☐	☐
Test Code 38/EPA Method 2021						
	<i>Daphnia pulex</i> - MHSF 25°C	LC50	☐	☐	☐	☐
Test Code 42/EPA Method 2007						
	Mysid (<i>Mysidopsis bahia</i>) 25°C	LC50	☐	☐	☐	☐
Test Code 43/EPA Method 1007						
	Mysid (<i>Mysidopsis bahia</i>)	NOEC SURVIVAL	☐	☐	☐	☐
	Mysid (<i>Mysidopsis bahia</i>)	IC25 (ON) GROWTH	☐	☐	☐	☐
	Mysid (<i>Mysidopsis bahia</i>)	NOEC (ON) GROWTH	☐	☐	☐	☐
Test Code 44/EPA Method 2006						
	Inland silverside (<i>Menidia beryllina</i>) 25°C	LC50	☐	☐	☐	☐
Test Code 45/EPA Method 1006						
	Inland silverside (<i>Menidia beryllina</i>)	NOEC SURVIVAL	☐	☐	☐	☐
	Inland silverside (<i>Menidia beryllina</i>)	IC25 (ON) GROWTH	☐	☐	☐	☐
	Inland silverside (<i>Menidia beryllina</i>)	NOEC (ON) GROWTH	☐	☐	☐	☐
Test Code 46/EPA Method 2004						
	Sheepshead minnow (<i>Cyprinodon variegatus</i>) 25°C	LC50	☐	☐	☐	☐
Test Code 47/EPA Method 1004						
	Sheepshead minnow (<i>Cyprinodon variegatus</i>)	NOEC SURVIVAL	☐	☐	☐	☐
	Sheepshead minnow (<i>Cyprinodon variegatus</i>)	IC25 (ON) GROWTH	☐	☐	☐	☐
	Sheepshead minnow (<i>Cyprinodon variegatus</i>)	NOEC (ON) GROWTH	☐	☐	☐	☐

Signature _____ Date _____

Permittee name

Cadillac WWTP

State

MI

NPDES Permit No.

0000757

EPA Lab Code

MI 00041

Chemistry/Microbiology Analyte Checklist

DMR-QA Study 31

Analyte Test	Test Required	Laboratory's Graded Result		Analyte determined by state-certified laboratory*
		Acceptable	Not Acceptable (Corrective Action Required)	
Microbiology				
<i>E. coli</i>	☐	☐	☐	☐
Fecal Coliform, MF or MPN	☐	☐	☐	☐
Total Coliform, MF or MPN	☐	☐	☐	☐
Trace Metals				
Aluminum	☐	☐	☐	☐
Antimony	☐	☐	☐	☐
Arsenic	☐	☐	☐	☐
Barium	☐	☐	☐	☐
Beryllium	☐	☐	☐	☐
Cadmium	☐	☐	☐	☐
Chromium, total	☐	☐	☐	☐
Chromium, hexavalent	☐	☐	☐	☐
Cobalt	☐	☐	☐	☐
Copper	☐	☐	☐	☐
Iron	☐	☐	☐	☐
Lead	☐	☐	☐	☐
Manganese	☐	☐	☐	☐
Mercury	☐	☐	☐	☐
Mercury (Low Level)	☒	☒	☐	☐
Molybdenum	☐	☐	☐	☐
Nickel	☐	☐	☐	☐
Selenium	☐	☐	☐	☐
Silver	☐	☐	☐	☐
Thallium	☐	☐	☐	☐
Vanadium	☐	☐	☐	☐
Zinc	☐	☐	☐	☐
Demands				
5-day BOD	☐	☐	☐	☐
5-day Carbonaceous BOD	☐	☐	☐	☐
COD	☐	☐	☐	☐
TOC	☐	☐	☐	☐
Minerals				
Alkalinity, total (CaCO ₃)	☐	☐	☐	☐
Chloride	☐	☐	☐	☐
Fluoride	☐	☐	☐	☐
Hardness, total (CaCO ₃)	☐	☐	☐	☐
Specific conductance (25°C)	☐	☐	☐	☐
Sulfate	☐	☐	☐	☐
Total Dissolved Solids (180°C)	☐	☐	☐	☐
Nutrients				
Ammonia as N	☐	☐	☐	☐
Nitrate as N	☐	☐	☐	☐
Nitrite as N	☐	☐	☐	☐
Orthophosphate as P	☐	☐	☐	☐
Total Kjeldahl-Nitrogen as N	☐	☐	☐	☐
Total Phosphorus as P	☐	☐	☐	☐
Misc. Analytes				
Non-Filterable Residue (TSS)	☐	☐	☐	☐
Oil and Grease	☐	☐	☐	☐
pH	☐	☐	☐	☐
Total Cyanide	☐	☐	☐	☐
Total Phenolics (4-AAP)	☐	☐	☐	☐
Total Residual Chlorine	☐	☐	☐	☐
Total Residual Chlorine (Low Level)	☐	☐	☐	☐
Settleable Solids	☐	☐	☐	☐
Turbidity	☐	☐	☐	☐

Signature

Date



Cadillac WWTP
Final Limits
For Study DMR-QA 31

LABORATORY INFORMATION

Cadillac WWTP
200 N. Lake St.
Cadillac, MI 49601

Customer ID: C454401
Fax: (231) 775-2876
Email: lab@cadillac-mi.net
Phone: (231) 775-2368

USEPA ID: MI00082
Contact: Amy Vail
Contact Title: Lab Manager
Approval Date: 6/15/2011

REGULATORY AGENCY INFORMATION

There are no agencies listed to receive reports.

THIRD PARTY INFORMATION

There are no third parties listed to receive reports.

Study DMRQA-31
Exception Report

This exception report shows only those analytes with a performance evaluation of 'Not Acceptable'.

Complex Nutrients (cat# 579)

Analyte	Units	Reported Value	Assigned Value	Acceptance Limits	Not Acceptable Evaluation	Method Description
Total phosphorus as P	mg/L	8.51	6.65	5.48 - 7.88	Not Acceptable	SM4500P E online

Study DMR-QA 31
Complete Final Report

Hardness (cat# 580)

Analyte	Units	Reported Value	Assigned Value	Acceptance Limits	Performance Evaluation	Method Description
Non-Filterable Residue (TSS)	mg/L	43.8	46.2	35.4 - 53.2	Acceptable	SM2540D online
Calcium	mg/L		29.7	26.3 - 33.9	Not Reported	
Magnesium	mg/L		12.1	10.3 - 13.9	Not Reported	
Calcium Hardness as CaCO3	mg/L		74.2	65.8 - 84.6	Not Reported	
Total Hardness as CaCO3	mg/L		124	108 - 142	Not Reported	

pH (cat# 577)

Analyte	Units	Reported Value	Assigned Value	Acceptance Limits	Performance Evaluation	Method Description
pH	S.U.	7.63	7.65	7.45 - 7.85	Acceptable	SM4500H+ B online

Simple Nutrients (cat# 584)

Analyte	Units	Reported Value	Assigned Value	Acceptance Limits	Performance Evaluation	Method Description
Ammonia as N	mg/L	12.8	11.8	8.76 - 14.7	Acceptable	SM4500NH3 D online
Nitrate + Nitrite as N	mg/L		4.23	3.44 - 4.93	Not Reported	
Nitrate as N	mg/L		4.23	3.29 - 5.12	Not Reported	
ortho-Phosphate as P	mg/L		3.57	2.93 - 4.24	Not Reported	

Complex Nutrients (cat# 579)

Analyte	Units	Reported Value	Assigned Value	Acceptance Limits	Performance Evaluation	Method Description
Total Kjeldahl Nitrogen	mg/L		19.3	12.7 - 24.9	Not Reported	
Total phosphorus as P	mg/L	8.51	6.65	5.48 - 7.88	Not Acceptable	SM4500P E online

Reported Values Last Update 7/11/2011

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Study DMR-QA 31
Complete Final Report

Demand (cat# 578)

Analyte	Units	Reported Value	Assigned Value	Acceptance Limits	Performance Evaluation	Method Description
BOD	mg/L		75.3	38.0 - 113	Not Reported	
CBOD	mg/L	71.2	64.8	29.1 - 100	Acceptable	SM5210B online
COD	mg/L		122	92.0 - 141	Not Reported	
TOC	mg/L		48.1	40.1 - 55.5	Not Reported	

WasteWatR™ Coliform MicrobE™ (cat# 576)

Analyte	Units	Reported Value	Assigned Value	Acceptance Limits	Performance Evaluation	Method Description
Total Coliforms (MF)	CFU/100mL		594	56.0 - 6270	Not Reported	
Fecal Coliforms (MF)	CFU/100mL	665	582	98.0 - 3460	Acceptable	SM9222D m FC online
E.coli (MF)	CFU/100mL		642	99.0 - 4160	Not Reported	
Total Coliforms (MPN)	MPN/100mL		1560	573 - 4240	Not Reported	
Fecal Coliforms (MPN)	MPN/100mL		1210	246 - 5970	Not Reported	
E.coli (MPN)	MPN/100mL		1460	501 - 4250	Not Reported	

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WS-175 Final Complete Report

Amy Jean Vail
Laboratory Manager
City of Cadillac WWTP Lab
200 N. Lake St.
Laboratory
Cadillac, MI 49601
(231) 775-2368

EPA ID: Not Reported
ERA Customer Number: C374701
Report Issued: 04/13/11
Study Dates: 02/07/11 - 03/24/11

Anal. No.	Analyte	Units	Reported Value	Assigned Value	Acceptance Limits	Performance Evaluation	Method Description
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WS Metals (cat# 590)

1000	Aluminum	µg/L		2010	1730 - 2220	Not Reported	
0140	Antimony	µg/L		37.2	26.0 - 48.4	Not Reported	
0001	Arsenic	µg/L		28.8	20.2 - 37.4	Not Reported	
0002	Barium	µg/L		845	718 - 972	Not Reported	
0141	Beryllium	µg/L		8.84	7.51 - 10.2	Not Reported	
0226	Boron	µg/L		1770	1540 - 1960	Not Reported	
0003	Cadmium	µg/L		15.5	12.4 - 18.6	Not Reported	
0004	Chromium	µg/L		191	162 - 220	Not Reported	
0091	Copper	µg/L	403	451	406 - 496	Not Acceptable	SM3111B
1070	Iron	µg/L		1130	1010 - 1240	Not Reported	
0005	Lead	µg/L		55.1	38.6 - 71.6	Not Reported	
0236	Manganese	µg/L		571	514 - 628	Not Reported	
0237	Molybdenum	µg/L		66.4	57.3 - 73.9	Not Reported	
0142	Nickel	µg/L	89	75.9	64.5 - 87.3	Not Acceptable	SM3111B
0007	Selenium	µg/L		74.0	59.2 - 88.8	Not Reported	
1150	Silver	µg/L		181	160 - 200	Not Reported	
0143	Thallium	µg/L		5.16	3.61 - 6.71	Not Reported	
1185	Vanadium	µg/L		546	491 - 601	Not Reported	
0239	Zinc	µg/L	617	604	544 - 664	Acceptable	SM3111B

WS Metals (cat# 590)

1000	Aluminum	µg/L		2010	1730 - 2220	Not Reported	
0140	Antimony	µg/L		37.2	26.0 - 48.4	Not Reported	
0001	Arsenic	µg/L		28.8	20.2 - 37.4	Not Reported	
0002	Barium	µg/L		845	718 - 972	Not Reported	
0141	Beryllium	µg/L		8.84	7.51 - 10.2	Not Reported	
0226	Boron	µg/L		1770	1540 - 1960	Not Reported	
0003	Cadmium	µg/L	30.0	15.5	12.4 - 18.6	Not Acceptable	SM3113B
0004	Chromium	µg/L	188	191	162 - 220	Acceptable	SM3113B
0091	Copper	µg/L		451	406 - 496	Not Reported	
1070	Iron	µg/L		1130	1010 - 1240	Not Reported	
0005	Lead	µg/L	54.4	55.1	38.6 - 71.6	Acceptable	SM3113B
0236	Manganese	µg/L		571	514 - 628	Not Reported	
0237	Molybdenum	µg/L	52.1	66.4	57.3 - 73.9	Not Acceptable	SM3113B
0142	Nickel	µg/L		75.9	64.5 - 87.3	Not Reported	
0007	Selenium	µg/L	66.4	74.0	59.2 - 88.8	Acceptable	SM3113B
1150	Silver	µg/L	155	181	160 - 200	Not Acceptable	SM3113B
0143	Thallium	µg/L		5.16	3.61 - 6.71	Not Reported	
1185	Vanadium	µg/L		546	491 - 601	Not Reported	
0239	Zinc	µg/L		604	544 - 664	Not Reported	

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All analytes are included in ERA's A2LA accreditation. Lab Code: 1539-01

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A Waters Company

Study: **DMRQA-31**

NPDES Permit #: **MI0020257**

Laboratory Name: **Cadillac WWTP**

Inorganic Results





A Waters Company

DMRQA-31 Final Complete Report

NPDES Permit #: MI0020257
Permit Holder: Amy Vail
Lab Manager
Cadillac WWTP
1121 Plett Rd.
Cadillac, MI 49601
(231) 775-2368

ERA Customer Number: C454401
Report Issued: 07/22/11
Study Dates: 03/14/11 - 07/01/11

Analyte	Performance Evaluation	Reported Value	Assigned Value	Acceptance Limits	Units	Method Description	USEPA Lab Code	Study
Low-Level Mercury								
Low Level Mercury	Acceptable	86.5	82.2	63.3 - 100	ng/L	EPA 1631B	MI00041	DMRQA31





A Waters Company

Study: **DMRQA-31**

NPDES Permit #: **MI0020257**

Laboratory Name: **Cadillac WWTP**

WET Results





A Waters Company

DMRQA-31 Final Complete Report

NPDES Permit #: MI0020257
Permit Holder: Amy Vail
Lab Manager
Cadillac WWTP
1121 Plett Rd.
Cadillac, MI 49601
(231) 775-2368

ERA Customer Number: C454401
Report Issued: 07/22/11
Study Dates: 03/14/11 - 07/01/11

Test End Point	Performance Evaluation	Reported Value %	Assigned Value %	Acceptance Limits %	Method Description	USEPA Lab Code	Study
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Ceriodaphnia dubia (Test Code 19)
48Hr., Acute Renewal, 25° C, MHSF
Potassium chloride

LC50	Acceptable	52.1	40.4	10.0 - 70.8	EPA 2002.0	MI01187	DMRQA31
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Ceriodaphnia dubia (Test Code 21)
7-day Short term Chronic, Daily Renewal, MHSF
Potassium chloride

IC25 Reproduction	Acceptable	28.4	25.2	12.2 - 38.1	EPA 1002.0	MI01187	DMRQA31
NOEC Reproduction	Acceptable	25.0	25.0	12.5 - 50.0	EPA 1002.0	MI01187	DMRQA31
NOEC Survival	Acceptable	25.0	25.0	12.5 - 50.0	EPA 1002.0	MI01187	DMRQA31

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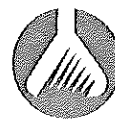
AUG 05 2011

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USEPA DMR-QA 31

NPDES PERMITTEE DATA REPORT FORM



ERA

A Waters Company

USEPA NPDES MI0020257

Permit #:

Permit Ext:

Permittee Name: Cadillac WWTP

Facility Address: 1121 Pielt Rd.

City: Cadillac

State: MI

Postal Code: 49601

Phone Number: (231) 775-2368

Fax Number: (231) 775-2876

E-mail address: lab@cadillac-mi.net

For DMR-QA Study 31, conducted in 2011, the Permittee ensured that their laboratory(s) performing the required analyses:

Received PT Samples

Yes ☐ No ☐

Submitted Complete and Accurate Data
by July 1, 2011

Yes ☐ No ☐

Received a Graded Report by July
22, 2011

Yes ☐ No ☐

Certification by Permit Holder or Authorized Representative
(as per 40 C.F.R. Section 122.22)

I certify under penalty of law that this document and all attachments were prepared under my direction or supervision in accordance with a system designed to assure that qualified personnel properly gather and evaluate the information submitted. Based on my inquiry of the person or persons who manage the system, or those persons directly responsible for gathering the information, the information submitted is, to the best of my knowledge and belief, true, accurate and complete. Each reported value was produced from a single analytical run using the analytical system that routinely performs these analyses to produce compliance monitoring data required under our National Pollutant Discharge Elimination System (NPDES) Permit. Neither I nor any of my subordinates compared our results from independent analyses conducted by us or any other laboratory before we reported our results to the USEPA. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment for knowing violations.

Name of

Certifying Official: Amy Vail

Title: Lab Manager

Signature:

Date Signed:

/ / 2011

Mailing Address: 200 N Lake Street

(enter only if different
from address above)

City: Cadillac

State: MI

Postal Code: 49601

Phone Number: (231) 775-2368

E-mail address: lab@cadillac-mi.net

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United States

ENVIRONMENTAL PROTECTION AGENCY



Laboratory Performance Evaluation Laboratory DMR-QA Evaluation Study 31

USEPA NPDES Permit #: MI0020257

Permit Ext:

Identification of all CHEM, MICRO and TOX laboratories who did analyses for this permit

Name of Laboratory	Address of Laboratory	U.S. EPA Lab Code	Lab Analysis Check box(es) that apply			Lab Type
			CHEM	MICRO	TOX	
Kar Labs	4425 Manchester Rd. Kalamazoo, MI 49001	MI00041	☒	☐	☐	
Paragon Laboratories, Inc.	12649 Richfield Ct Livonia, MI 48150	MI01187	☐	☐	☒	

Lab Types: C = Commerical F = Federal G = Local Government I = Industrial O = Other S = State



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Permittee Name: Cadillac WWTP

Permit Number: MI0020257

EPA Lab Code: MI00041

Chemistry/Microbiology Analyte Checklist

DMRQA Study 31

Analyte Test / Method	Test Required	Laboratory's Graded Result	
		Acceptable	Not Acceptable (Corrective Action Required)
Minerals Alkalinity as CaCO ₃ Chloride Conductivity at 25°C Fluoride Potassium Sodium Sulfate Total Dissolved Solids at 180°C Total Solids at 105°C			
Hardness Non-Filterable Residue (TSS) Calcium Magnesium Calcium Hardness as CaCO ₃ Total Hardness as CaCO ₃			
pH pH			
Settleable Solids Settleable Solids			
Solids Concentrate Non-Filterable Residue (TSS) Total Dissolved Solids at 180°C Total Solids at 105°C			
Solids Non-Filterable Residue (TSS) Total Dissolved Solids at 180°C Total Solids at 105°C			
Simple Nutrients Ammonia as N Nitrate + Nitrite as N Nitrate as N ortho-Phosphate as P			
Complex Nutrients Total Kjeldahl Nitrogen Total phosphorus as P			
Nitrite Nitrite as N			
Demand BOD			



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Permit Number: MI0020257

EPA Lab Code: MI00041

Chemistry/Microbiology Analyte Checklist

DMRQA Study 31

Analyte Test / Method	Test Required	Laboratory's Graded Result	
		Acceptable	Not Acceptable (Corrective Action Required)
CBOD COD TOC			
<u>Oil & Grease Concentrate</u> Oil & Grease (Gravimetric)			
<u>Oil & Grease</u> Oil & Grease (Gravimetric) Oil & Grease (Infrared)			
<u>Trace Metals</u> Aluminum Antimony Arsenic Barium Beryllium Boron Cadmium Chromium Cobalt Copper Iron Lead Manganese Molybdenum Nickel Selenium Silver Strontium Thallium Vanadium Zinc			
<u>Mercury</u> Mercury			
<u>Low-Level Mercury</u> Low Level Mercury EPA 1631B	☐	☒	☐
<u>Hexavalent Chromium</u> Hexavalent Chromium			
<u>Turbidity</u> Turbidity			
<u>Total Cyanide</u> Cyanide, total			

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Permittee Name: Cadillac WWTP

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EPA Lab Code: MI00041

Chemistry/Microbiology Analyte Checklist

DMRQA Study 31

Analyte Test / Method	Test Required	Laboratory's Graded Result	
		Acceptable	Not Acceptable (Corrective Action Required)
<u>Total Phenolics (4-AAP)</u> Phenolics, total			
<u>Total Residual Chlorine</u> Total Residual Chlorine			
<u>Low-Level Total Residual Chlorine</u> Low Level Total Residual Chlorine			
<u>WasteWatR™ Coliform MicrobE™</u> Total Coliforms (MF) Fecal Coliforms (MF) E.coli (MF) Total Coliforms (MPN) Fecal Coliforms (MPN) E.coli (MPN)			

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Print Name

Date

Signature/Title

Use a separate checklist for EACH lab used



A Waters Company

Permittee Name: Cadillac WWTP

Permit Number: MI0020257

EPA Lab Code: MI01187

WET Organisms/Test Conditions/End Points Checklist

DMRQA Study 31

Analyte Number	Organisms / Conditions	End Points	Test Required	Laboratory's Graded Result	
				Acceptable	Not Acceptable (Corrective Action Required)
Fathead minnow (Test Code 13)					
0754	48Hr., Acute, Non-Renewal, 25° C, MHSF	LC50	☐	☐	☐
Fathead minnow (Test Code 14)					
0755	48Hr., Acute, Non-Renewal, 25° C, 20% DMW	LC50	☐	☐	☐
Fathead minnow (Test Code 15)					
0808	7-day Short term Chronic, Daily Renewal, MHSF	IC25 (ON) Growth	☐	☐	☐
0810	7-day Short term Chronic, Daily Renewal, MHSF	NOEC (ON) Growth	☐	☐	☐
0756	7-day Short term Chronic, Daily Renewal, MHSF	NOEC Survival	☐	☐	☐
Fathead minnow (Test Code 16)					
0812	7-day Short term Chronic, Daily Renewal, 20% DMW	IC25 (ON) Growth	☐	☐	☐
0814	7-day Short term Chronic, Daily Renewal, 20% DMW	NOEC (ON) Growth	☐	☐	☐
0759	7-day Short term Chronic, Daily Renewal, 20% DMW	NOEC Survival	☐	☐	☐
Ceriodaphnia dubia (Test Code 19)					
0764	48Hr., Acute Renewal, 25° C, MHSF	LC50	☐	☐	☐
0764	48Hr., Acute Renewal, 25° C, MHSF	LC50	☐	☒	☐
Ceriodaphnia dubia (Test Code 20)					
0765	48Hr., Acute Renewal, 25° C, 20% DMW	LC50	☐	☐	☐
Ceriodaphnia dubia (Test Code 21)					
0767	7-day Short term Chronic, Daily Renewal, MHSF	IC25 Reproduction	☐	☐	☐
0767	7-day Short term Chronic, Daily Renewal, MHSF	IC25 Reproduction	☐	☒	☐
0768	7-day Short term Chronic, Daily Renewal, MHSF	NOEC Reproduction	☐	☐	☐
0768	7-day Short term Chronic, Daily Renewal, MHSF	NOEC Reproduction	☐	☒	☐
0766	7-day Short term Chronic, Daily Renewal, MHSF	NOEC Survival	☐	☐	☐
0766	7-day Short term Chronic, Daily Renewal, MHSF	NOEC Survival	☐	☒	☐
Ceriodaphnia dubia (Test Code 22)					
0770	7-day Short term Chronic, Daily Renewal, 20% DMW	IC25 Reproduction	☐	☐	☐
0771	7-day Short term Chronic, Daily Renewal, 20% DMW	NOEC Reproduction	☐	☐	☐
0769	7-day Short term Chronic, Daily Renewal, 20% DMW	NOEC Survival	☐	☐	☐
Daphnia magna (Test Code 32)					
0788	48Hr., Acute, Non-Renewal, 25° C, MHSF	LC50	☐	☐	☐
Daphnia pulex (Test Code 38)					
0794	48Hr., Acute, Non-Renewal, 25° C, MHSF	LC50	☐	☐	☐
Mysid (Test Code 42)					
0798	48Hr., Acute, Non-Renewal, 25° C, 40 FSW	LC50	☐	☐	☐
Inland Silverside (Test Code 45)					
0825	7-day Short term Chronic, Daily Renewal, 40 FSW	IC25 (ON) Growth	☐	☐	☐
0826	7-day Short term Chronic, Daily Renewal, 40 FSW	NOEC (ON) Growth	☐	☐	☐
0824	7-day Short term Chronic, Daily Renewal, 40 FSW	NOEC Survival	☐	☐	☐



A Waters Company

Permittee Name: Cadillac WWTP

Permit Number: MI0020257

EPA Lab Code: MI01187

WET Organisms/Test Conditions/End Points Checklist

DMRQA Study 31

Analyte Number	Organisms / Conditions	End Points	Test Required	Laboratory's Graded Result	
				Acceptable	Not Acceptable (Corrective Action Required)
Mysid (Test Code 43)					
0816	7-day Short term Chronic, Daily Renewal, 40 FSW	IC25 (ON) Growth	☐	☐	☐
0818	7-day Short term Chronic, Daily Renewal, 40 FSW	NOEC (ON) Growth	☐	☐	☐
0799	7-day Short term Chronic, Daily Renewal, 40 FSW	NOEC Survival	☐	☐	☐
Inland silverside (Test Code 44)					
0803	48Hr., Acute, Non-Renewal, 25° C, 40 FSW	LC50	☐	☐	☐
Sheepshead minnow (Test Code 46)					
0804	48Hr., Acute, Non-Renewal, 25° C, 40 FSW	LC50	☐	☐	☐
Sheepshead minnow (Test Code 47)					
0820	7-day Short term Chronic, Daily Renewal, 40 FSW	IC25 (ON) Growth	☐	☐	☐
0822	7-day Short term Chronic, Daily Renewal, 40 FSW	NOEC (ON) Growth	☐	☐	☐
0805	7-day Short term Chronic, Daily Renewal, 40 FSW	NOEC Survival	☐	☐	☐

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Print Name

Date

Signature/Title

Use a separate checklist for EACH lab used



A Waters Company

Study: **DMRQA-31**

NPDES Permit #: **MI0020257**

Laboratory Name: **Cadillac WWTP**

Inorganic Results

WATER RESOURCES
DIVISION

DEC - 5 2011

CADILLAC DISTRICT





A Waters Company

DMRQA-31 Final Complete Report

NPDES Permit #: MI0020257
Permit Holder: Amy Vail
Lab Manager
Cadillac WWTP
1121 Plett Rd.
Cadillac, MI 49601
(231) 775-2368

ERA Customer Number: C454401
Report Issued: 07/22/11
Study Dates: 03/14/11 - 07/01/11

Analyte	Performance Evaluation	Reported Value	Assigned Value	Acceptance Limits	Units	Method Description	USEPA Lab Code	Study
Low-Level Mercury								
Low Level Mercury	Acceptable	86.5	82.2	63.3 - 100	ng/L	EPA 1631B	MI00041	DMRQA31





A Waters Company

Study: **DMRQA-31**

NPDES Permit #: **MI0020257**

Laboratory Name: **Cadillac WWTP**

WET Results





A Waters Company

DMRQA-31 Final Complete Report

NPDES Permit #: MI0020257
Permit Holder: Amy Vail
Lab Manager
Cadillac WWTP
1121 Plett Rd.
Cadillac, MI 49601
(231) 775-2368

ERA Customer Number: C454401
Report Issued: 07/22/11
Study Dates: 03/14/11 - 07/01/11

Test End Point	Performance Evaluation	Reported Value %	Assigned Value %	Acceptance Limits %	Method Description	USEPA Lab Code	Study
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Ceriodaphnia dubia (Test Code 19)
48Hr., Acute Renewal, 25° C, MHSF
Potassium chloride

LC50	Acceptable	52.1	40.4	10.0 - 70.8	EPA 2002.0	MI01187	DMRQA31
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Ceriodaphnia dubia (Test Code 21)
7-day Short term Chronic, Daily Renewal, MHSF
Potassium chloride

IC25 Reproduction	Acceptable	28.4	25.2	12.2 - 38.1	EPA 1002.0	MI01187	DMRQA31
NOEC Reproduction	Acceptable	25.0	25.0	12.5 - 50.0	EPA 1002.0	MI01187	DMRQA31
NOEC Survival	Acceptable	25.0	25.0	12.5 - 50.0	EPA 1002.0	MI01187	DMRQA31





200 North Lake Street • Cadillac, Michigan 49601
231.775.0181 • fax 231.775.8755
www.cadillac-mi.net

M10020257

8/3/11

Chris

The Cadillac WWTP had one "Not Acceptable" reading on our DMR-QA 31 for the total phosphorus as P analyte. The upper limit was 7.88 mg/L and we reported 8.51 mg/L. I believe this error in reading was due to the lower dilutions of the sample that needed to be read to get readings in our calibration curve. I have gone back and looked at all the dilutions that were run and if we averaged them all we would have been in acceptable limits for the PT. However, I did not use all the dilutions we ran because the higher dilutions were not in our standard curve range and are not as reliable as those that fall in the curve range.

I don't feel that our method/procedure was the cause of this failure due to the fact that we ran a standard, blank, and QC sample with the DMR-QA 31 sample and they all came in with 91.6-102.3% recovery.

I feel confident that this "Not Acceptable" result is not an indication of continuous failures and inaccurate data being reported on our monthly DMR. We do quarterly QC checks which also include a lab split with a nearby lab. This is the first time we've had any value out of range since I have taken over the lab five years ago. If the lower dilutions were the issue, that should not affect our DMR results as we use fairly higher dilutions for our raw wastewater and don't dilute the effluent at all.

I have had another PT test performed using more samples run at lower dilutions. I will forward you the results of this test when the study ends on August 25, 2011.

If you have any questions or concerns, please feel free to contact me at (231)775-2368.

Thank you,

A handwritten signature in cursive script, appearing to read "Amy Vail".

Amy Vail

City of Cadillac
Lab Manager

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